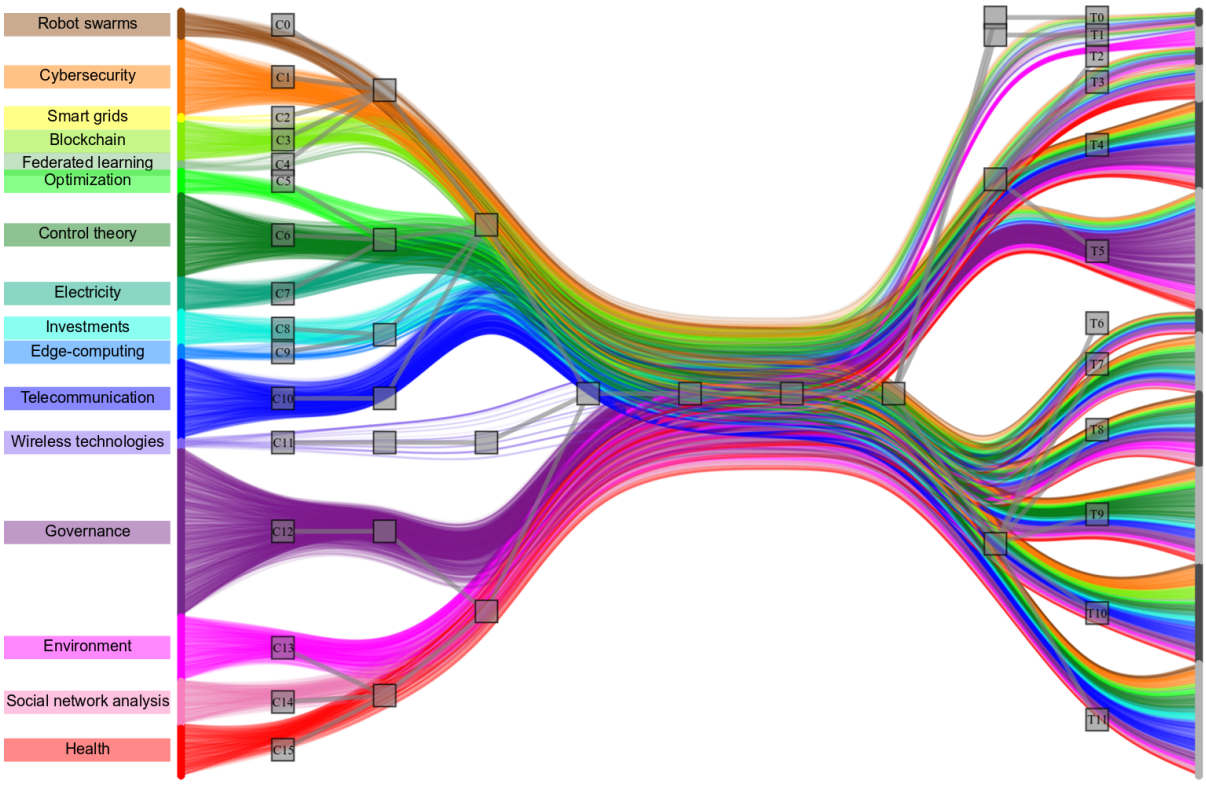

T0 (control, optimal, access, decentralised, cooperative, method, robust, peer, controller, multiple)
T1 (based, management, study, service, case, strategy, evaluation, review, community, towards)
T2 (economic, urban, low, heterogeneous, web, organization, environmental, city, uncertain, business)
T3 (technology, flow, descentralización, em, décentralisation, technologies, na, descentralização, saúde, brasil)
T4 (power, energy, centralized, information, supply, state, chain, coordination, planning, impact)
T5 (performance, process, centralization, treatment, use, industrial, driven, cancer, project, centralizers)
T6 (decentralization, local, development, health, governance, public, social, fiscal, government, china)
T7 (data, blockchain, sensor, mobile, resource, framework, smart, architecture, grid, allocation)
T8 (network, distributed, networks, multi, analysis, agent, algorithm, learning, dynamic, policy)
T9 (decentralized, system, using, systems, approach, model, wireless, design, time, large)
T10 (dual, exchange, loop, trading, closed, sensitive, reverse, remanufacturing, transactive, commodity)
T11 (new, demand, key, evolution, autonomy, action, identity, transformation, authority, motion)

Figure S6: **Visualization of the results of the hSBM algorithm.** Bipartite network of documents and words, where nodes and links are colored based on the document cluster (at hierarchical level 3) they start from. Multiple instances of the same word are considered, one for each document the word is in. Word nodes within a topic at level 3 are ordered as the document clusters for visualization purposes. On top of the bipartite network, the hierarchy of topics and document clusters are represented on the word and document side respectively, shown as a network of square gray nodes. On the bottom of the figure, the legend with the 10 most frequent words of each topic is displayed.

S3 The importance of Governance and Blockchain in the history of *(de)centralization*

In this section we report some additional and complementary plots to show the importance of the Blockchain and Governance clusters in the academic literature on *(de)centralization*. In Fig. S7(a) we show the number of papers in time for three groups of documents: Blockchain, Governance and the whole dataset. Moreover, in Fig. S7(b) we plot the rank of each cluster in terms of yearly number of papers, highlighting Blockchain and Governance. Both figures clearly show how Governance has been the most productive cluster in the literature for a long time, only to be replaced by Blockchain in recent years.

In Fig. S8 we look at knowledge flows, plotting their value instead of their rank (as done in Fig. 3 in the main text). We plot the time evolution of $K_{a\to\bullet}(Y)$, where a is Governance or Blockchain, in comparison to $K_{\bullet\to\bullet}(Y)$ in Fig. S8(a). This shows the average influence that the cluster in a certain year has towards all clusters in the future. We similarly compare the average influence of all other clusters on the future of these two clusters looking at $K_{\bullet\to a}(Y)$ in Fig. S8(b). We can see from these figures how Governance has been increasingly important in influencing other clusters until the 1980s, while since after the 1990s it has had a lower knowledge flow than the average among all clusters, despite being the first cluster in terms of number of papers in all this time. The case of Blockchain is opposite: after 2013 it starts to have a much higher influence towards the other clusters compared to the average one over all clusters, while the other clusters after 2004 have had an average knowledge flow towards it.

In Fig. S9, we show the average knowledge flow of a cluster a to the future, while also distinguishing the influence on the cluster itself. Fig. S9(a) focuses on the comparison of knowledge flows between Governance and other clusters, while Fig. S9(b) repeats the analysis on Blockchain. Starting from the 1970s, we observe a substantial knowledge flow within the Governance cluster towards future Governance papers (blue line), consistently close to 1. In comparison, other clusters have had a lower impact on their future papers during this period (magenta line), with a more noticeable and increasing influence in recent years. Examining the knowledge flow from Governance to other clusters (red line), we find that it has consistently been low, peaking in the 1990s. However, after the year 2000, the impact on other clusters has vanished. Conversely, all other clusters have exerted an increasing influence on all clusters excluding themselves (gray line), indicating their higher impact on other areas and their more efficient interdisciplinary communication. Hence, while the number of papers in Governance has remained high, its impact has largely been confined to the same cluster after the 1990s. Conversely, the opposite trend is observed for other clusters, where their influence on other disciplines has progressively grown. Furthermore, the influence of other clusters on Governance (black line) has been negligible since the 1990s, reinforcing the isolation of the Governance cluster. In contrast, the Blockchain cluster, gaining significance in the 2000s (blue line in Fig. S9(b)), has received substantial knowledge flow from other clusters (black line), particularly during that period. This influx of knowledge has contributed to the emergence of blockchain technologies, which have subsequently made a significant impact on other clusters in recent years, as noticed by the high knowledge flow from Blockchain to other clusters (red line), thereby driving the recent scientific advancements in decentralization.

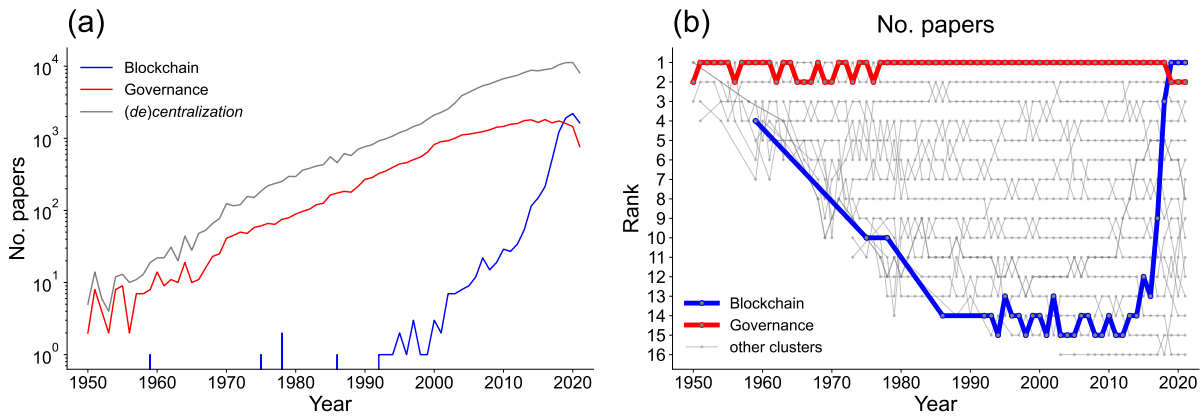

Figure S7: **Blockchain and Governance are the most productive clusters. (a)** Number of papers in time for the Blockchain cluster, the Governance cluster and the full *(de)centralization* dataset. **(b)** Rank by number or papers in time for different clusters, highlighting Blockchain and Governance. Both plots show the central role of these two clusters in the literature on *(de)centralization*, and how they have exchanged roles, with Blockchain becoming the most productive field in recent years.

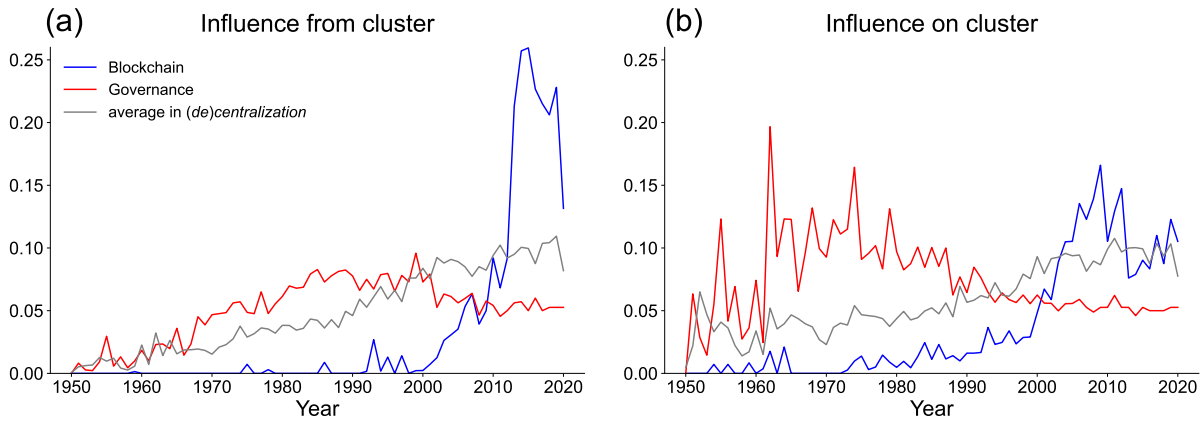

Figure S8: **Influence of Governance and Blockchain from and on other clusters in time.** Average knowledge flow towards other clusters in the future in time from a cluster **(a)** or to a cluster **(b)**, the latter being either Blockchain (in blue) or Governance (in red), while in gray the average over the *(de)centralization* dataset is plotted.

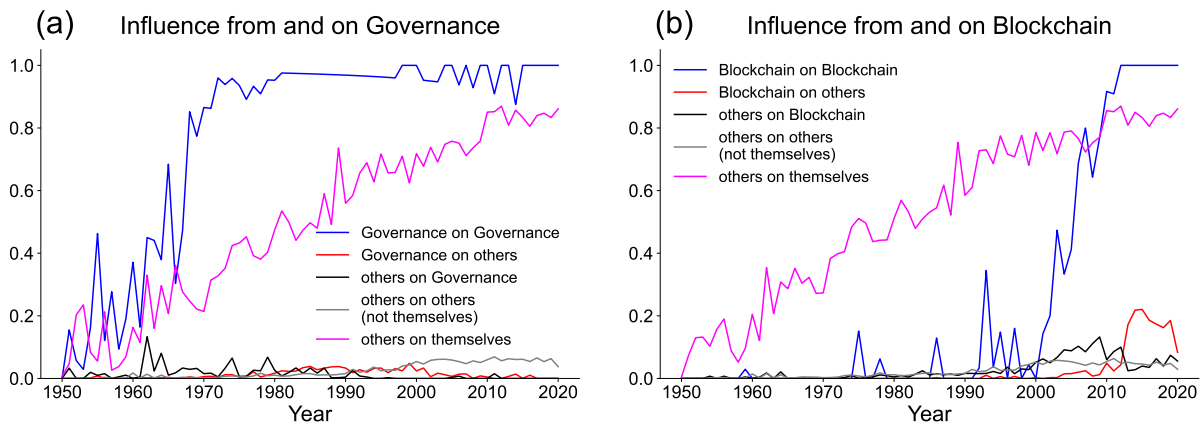

Figure S9: **Influence of Governance and Blockchain from and on other clusters in time.** Average knowledge flows from or to Governance (**(a)**) or Blockchain (**(b)**) compared to the other clusters, distinguishing between the knowledge flow to the same cluster or to all other ones.